

LESSON 2-10 • SESSION 1 • PRACTICE • FORM B

Mean or Median?

Learning target – I can choose the mean or the median to describe a data set based on its shape. • 6.SP.B.5d

REMEMBER

- An **outlier** pulls the **mean** toward it. The **median** (middle value) barely moves.
- **Skewed** data or an outlier: use the **median**.
Symmetric, no outlier: either works; use the **mean**.
- Mean goes with MAD. Median goes with IQR.

WORKED EXAMPLE

Five friends saved \$6, \$7, \$8, \$9 and \$50. Which center describes a typical friend?

Step 1 50 is an outlier.**Step 2** Median = 8. Mean = $80 \div 5 = 16$, more than four of the five values.**Step 3** The median is closer to most of the values.**Answer** The median, \$8**WORDS** outlier (*valor atípico*) • skewed (*asimétrico*) • symmetric (*simétrico*) • measure of center (*medida de centro*)**PRACTICE**

show your work • label units

1. Write the better measure of center: mean or median.

a. 30, 32, 34, 36, 38 _____

b. 30, 32, 34, 36, 700 _____

c. 5, 6, 7, 8, 9, 10 _____

d. 1, 45, 47, 49, 51 _____

2. **Trees in One Park.** Five trees are 14, 17, 19, 22 and 78 years old.

a. Outlier: _____

b. Median: _____ years

c. Mean: _____ years

d. Better center: _____

3. **Movie Runtimes.** Five runtimes are 92, 96, 99, 104 and 154 minutes.

Hint Put the values in order. The median is the middle one.

a. Median: _____ min

b. Mean: _____ min

c. Which measure is pulled up by the long film? _____

Show your work.

4. **Taxi Fares.** Five fares are \$10, \$13, \$15, \$17 and \$85.

a. Median: _____ dollars

b. Mean: _____ dollars

Show your work.

5. **Typical Value Claim.** For 40, 44, 47, 49, 180, Tyrell says the mean is a good typical value. Find the mean. Is he right?

Hint Count how many values sit near the mean.

Show your work.

Mean _____

6. **Salary Report.** Six salaries, in thousands of dollars, are 36, 38, 40, 42, 44 and 208.

Hint With six values, average the two middle values.

- a. Mean: _____
- b. Median: _____
- c. Which should a reporter publish? Why? _____

Show your work.

7. **No Outlier.** Five test scores are 70, 72, 74, 76 and 78. Find the mean and the median. Which would you report?

Hint Check how close the two measures land.

Show your work.

Mean, median _____

8. **Family Party.** Seven ages at a party are 6, 8, 10, 13, 15, 16 and 72.

- a. Median: _____ years
- b. Mean: _____ years
- c. Which center describes the group better? Which spread goes with it? _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. **Jonah's Work.** For 3, 4, 6, 8, 154, Jonah reported the mean, 35, as the typical value. Explain his mistake and give a better typical value.

TEST PRACTICE

choose the best answer

10. Roughly symmetric data with no outlier is best described by the –

- A range B IQR C outlier D mean

11. The mean naturally pairs with the –

- A IQR B median C MAD D range

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

19 M	37.5 X	40 Q	28 D	74 N	13 S	99 E	72 I	90 Z	41 A
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#2b	#3a	#4b	#5	#6b	#7	#8a
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CHECK YOURSELF

- ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____